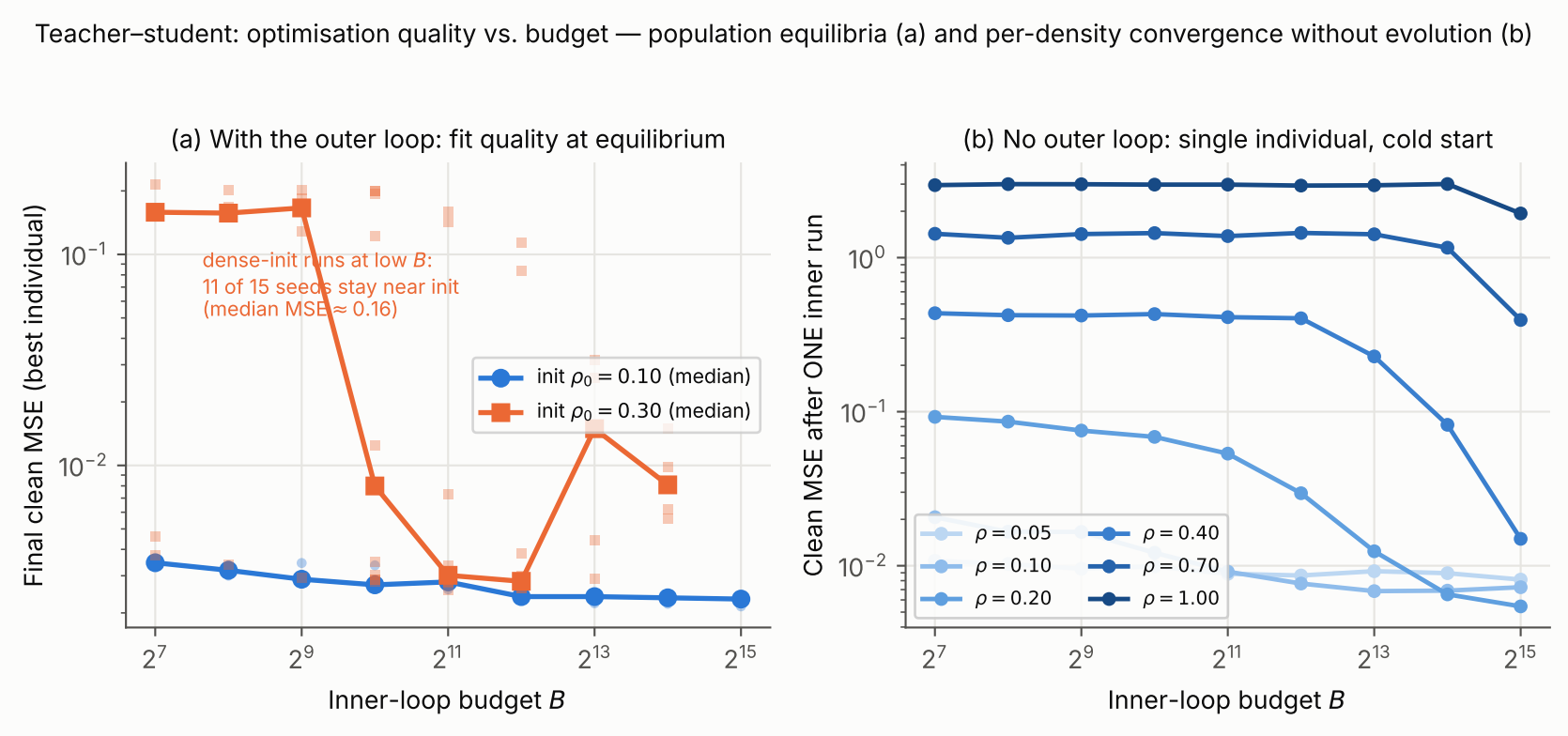
Teacher–student: optimisation quality vs. budget — population equilibria (a) and per-density convergence without evolution (b)
(a) With the outer loop: fit quality at equilibrium
Final clean MSE (best individual)
10^{-1}
10^{-2}
dense-init runs at low B:
11 of 15 seeds stay near init
(median MSE ≈ 0.16)
init $\rho_0 = 0.10$ (median)
init $\rho_0 = 0.30$ (median)
2^7
2^9
2^{11}
2^{13}
2^{15}
Inner-loop budget B
(b) No outer loop: single individual, cold start
Clean MSE after ONE inner run
10^{0}
10^{-1}
10^{-2}
$\rho = 0.05$
$\rho = 0.10$
$\rho = 0.20$
$\rho = 0.40$
$\rho = 0.70$
$\rho = 1.00$
2^7
2^9
2^{11}
2^{13}
2^{15}
Inner-loop budget B